

第15章 智能三维重建技术



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本次课程内容

1. SFM与SLAM

2. 从单幅图像重建

3. 基于NerF的技术

4. 3D Gaussian

5. 基于SDF的重建技术

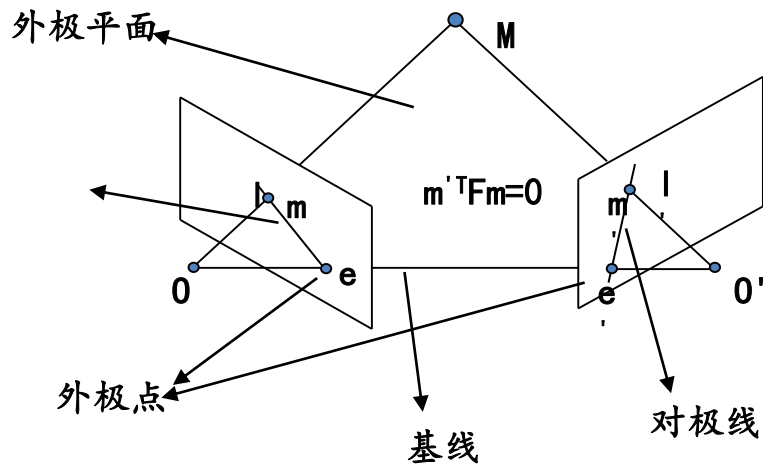
1. 结构恢复与SLAM

(1) SFM

SFM (Structure from Motion)，即从运动恢复结构，利用两幅图像恢复运动及摄像机的结构。

SFM是三维重建的基础路线之一。

极限几何是SFM问题模型基础



SFM解决措施

◆ 传统方法：

- 思路优化技术得到估计参数，根据摄像机成像模型
- 基本步骤：检测特征点、特征点匹配、优化恢复摄像机内部参数外部参数

◆ 智能方法：深度学习的优化技术

- 特点：摄像机的运动参数由网络回归得到

传统方法求解SFM方法

优化技术：先根据成像特性求取内部参数，再由本质矩阵，求取外部参数

先求内部参数矩阵：

根据：空间中的二次曲线的在图像平面的投影仍是二次曲线，是平面上的二次曲线

二次曲线方程 $X^T X = 0$ $X = \begin{pmatrix} x & y & z \end{pmatrix}^T$

再由小孔成像规律： $m \approx K \begin{pmatrix} R & T \end{pmatrix} \begin{pmatrix} X \\ 0 \end{pmatrix}$, $X \approx R^T K^{-1} m$ X代入上式得： $m^T K^{-T} K^{-1} m = 0$

对于给定正定矩阵 $C = K^{-T} K^{-1}$ ，则K（内部参数矩阵）可以通过Cholesky 分解唯一确定。

进一步求外部参数矩阵：

由两视图成像规律： $\begin{pmatrix} m'_i \end{pmatrix}^T K^{-T} [T]_x R K^{-1} m_i = 0$ $F \approx K^{-T} [T]_x R K^{-1}$ (1) (1) 中本质矩阵记为 $E = [T]_x R$

根据公式 (1)： m_i 及 m'_i 由两幅图像对应求得，再根据上步求得的摄像机内部参数K，可以求得本质矩阵 $E = [T]_x R$

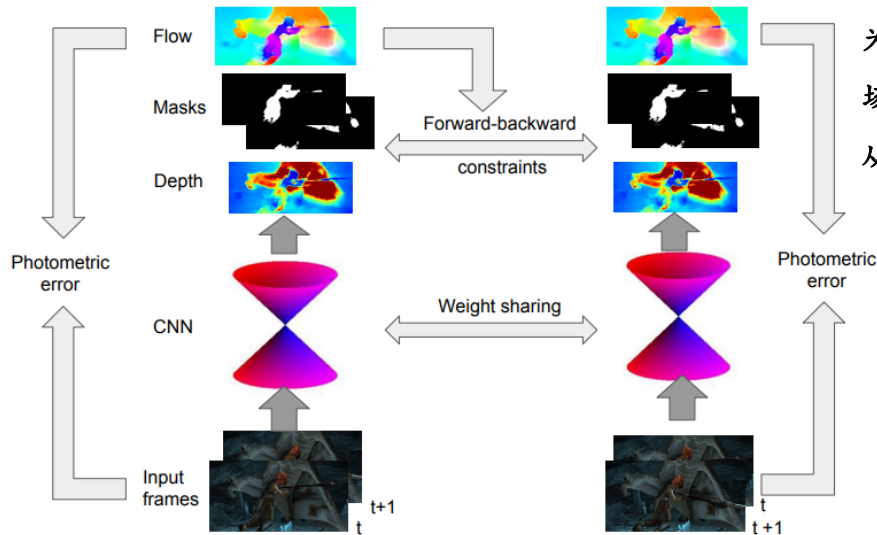
将 $E = [T]_x R$ 利用SVD分解，得到R和T矩阵

基于深度学习的结构恢复

实例： 从多幅图像重建—SfM-Net

- ◆ Sudheendra Vijayanarasimhan, Susanna Ricco, Cordelia Schmid, Rahul Sukthankar, Katerina Fragkiadaki: SfM-Net: Learning of Structure and Motion from Video. CoRR abs/1704.07804 (2017)
- ◆ SfM-Net, 对运动估计, 几何感知网络, 能够估计场景、目标深度、相机位姿、3D目标旋转和平移。
- ◆ 可以利用不同水平的监督
- ◆ 完全无监督（自监督）监督。
 - 自监督：利用重投影光度误差
 - 监督：相机运动
 - 监督：RGBD相机提供的深度信息
- ◆ 能将场景中的运动物体分离出来

前后一致性约束



SfM-Net: 给定一对帧作为输入，将帧到帧的像素运动分解为三维场景深度、三维摄像机的旋转和平移；将生成的3D场景流反向投影到二维光流中，并相应地warp，以将像素从一帧匹配到下一帧，利用前后一致性约束

Figure 1. SfM-Net: Given a pair of frames as input, our model decomposes frame-to-frame pixel motion into 3D scene depth, 3D camera rotation and translation, a set of motion masks and corresponding 3D rigid rotations and translations. It backprojects the resulting 3D scene flow into 2D optical flow and warps accordingly to match pixels from one frame to the next. Forward-backward consistency checks constrain the estimated depth.

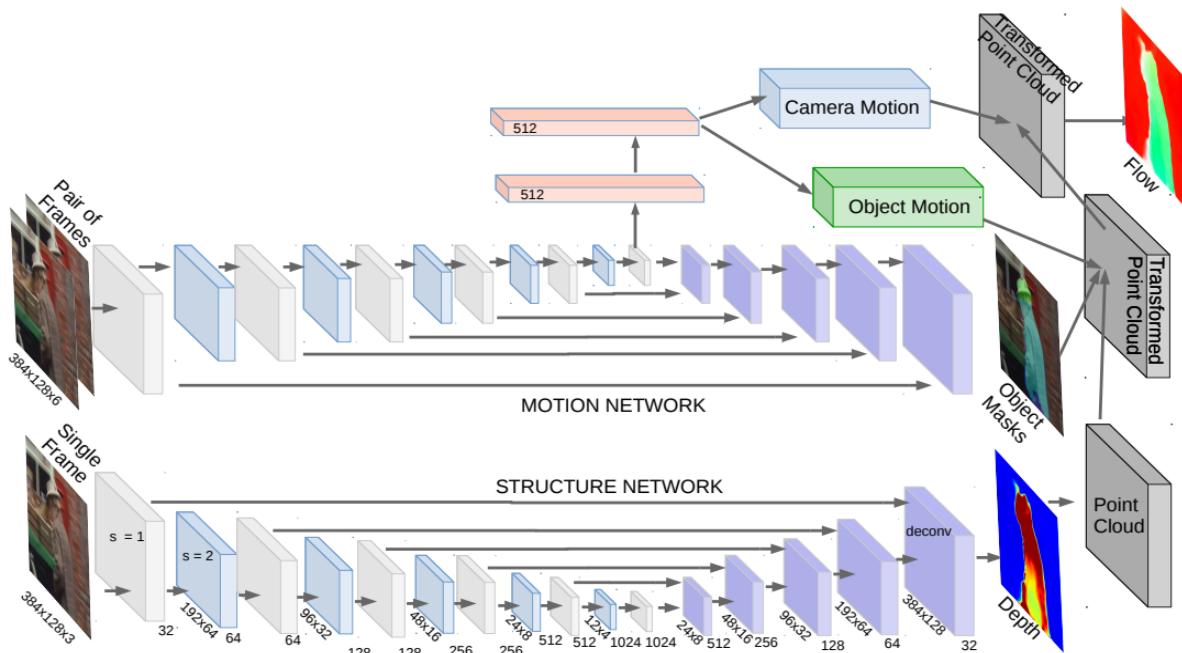


Figure 2. SfM-Net architecture. For each pair of consecutive frames I_t, I_{t+1} , a conv/deconv sub-network predicts depth d_t while another predicts a set of K segmentation masks m_t . The coarsest feature maps of the motion-mask encoder are further decoded through fully connected layers towards 3D rotations and translations for the camera and the K segmentations. The predicted depth is converted into a per frame point-cloud using estimated or known camera intrinsics. Then, it is transformed according to the predicted 3D scene flow, as composed by the 3D camera motion and independent 3D mask motions. Transformed 3D depth is projected back to the 2D next frame, and thus provides corresponding 2D optical flow fields. Differentiable backward warping maps frame I_{t+1} to I_t , and gradients are computed based on pixel errors. Forward-backward constraints are imposed by repeating this process for the inverted frame pair I_{t+1}, I_t and constraining the depths d_t and d_{t+1} to be consistent through the estimated scene motion.

无监督学习部分

- ◆ **Self-Supervision:**

计算光流，loss设计依据光照不变性假设

- ◆ **Spatial smoothness priors:**

使用L1范数惩罚，以保证空间的平滑性；深度预测学习，采用loss为二阶梯度的范数，以鼓励平稳地改变深度值。

- ◆ **Forward-backward consistency constraints:** 深度一致性约束的自监督策略



监督学习部分

- ◆ **Supervising depth:**

使用深度回归，进行深度监督

- ◆ **Supervising camera motion:** 相机真实运动轨迹对相机的位姿进行监督。

- ◆ **Supervising optical flow and object motion:**

使用L1 loss对光流训练进行监督

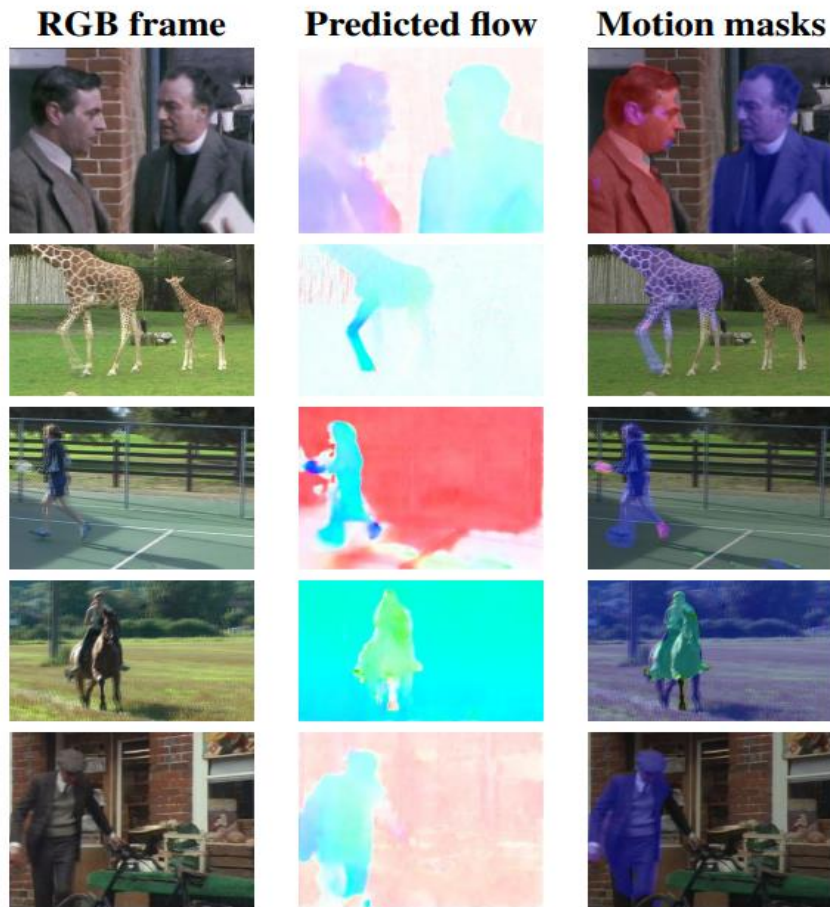


Figure 5. Motion segments computed from SfM-Net in MoSeg [4].
The model was trained in a fully unsupervised manner.

(2) SLAM介绍

- ◆ SLAM, Simultaneous localization and mapping (同步定位与建图)
- ◆ SLAM主要用于解决机器人在未知环境运动时的定位与地图构建问题

SLAM的典型应用领域——机器人定位导航

- ◆ SLAM地图建模。
- ◆ SLAM辅助机器人执行路径规划、自主探索、导航等任务。
- ◆ 机器人可以通过用SLAM算法结合激光雷达或者摄像头的方法，高效绘制室内地图，智能分析和规划扫地环境，智能导航

SLAM的典型应用领域——VR/AR

- ◆ 利用SLAM技术构建视觉效果更为真实的地图，针对当前视角渲染虚拟物体的叠加效果，使之更具真实感。
- ◆ VR/AR代表性产品中，微软Hololens、谷歌ProjectTango及MagicLeap都应用了SLAM作为视觉增强手段。

SLAM的典型应用领域——无人机领域

- ◆ 地图建模
- ◆ SLAM可以快速构建局部3D地图，并与地理信息系统（GIS）、视觉对象识别相结合，可以辅助无人机识别路障目标并自动避障
- ◆ Hovercamera无人机应用了SLAM技术

SLAM的典型应用领域——无人机领域

- ◆ 视觉里程计
- ◆ SLAM可以提供视觉里程计功能，并与GPS等其他定位方式相融合，满足无人驾驶精准定位的需求。
- ◆ 基于激光雷达技术的无人驾驶汽车，应用此技术

SLAM框架

- ◆ SLAM系统框架如图，一般分为五个模块，包括传感器数据、视觉里程计、后端优化、闭环检测和构图



视觉SLAM

- ◆ 视觉SLAM功能是获取外界信息的主要来源。
- ◆ 视觉SLAM从环境中获取海量信息
- ◆ SLAM在复杂动态场景中跟踪和预测场景中动态目标，如行人、车辆等
- ◆ 近年来，SLAM导航技术已取得了很大的发展，它将赋予机器人和其他智能体智能行为

深度学习结合SLAM的三个方向

- ◆ 传统SLAM几个模块：
 - 特征提取，特征匹配
 - 深度估计
 - 位姿估计
 - 重定位
 - 其他
- ◆ 智能SLAM，在传统技术基础上加入语义信息，包括图像语义分割及语义地图构建
- ◆ 端到端的SLAM：输入image输出action

传统SLAM与SFM方法区别

◆ SLAM与SFM区别：

- SFM 图像无序的、离线处理的，对处理时间要求不严格；SLAM图像有序的，在线处理的，对处理时间要求严格；
- SFM一般用在大的范围，如构建谷歌地图；SLAM中尺度较大也就达到几千米

传感器数据
(单目/双目
图像, 深度,
惯性等)



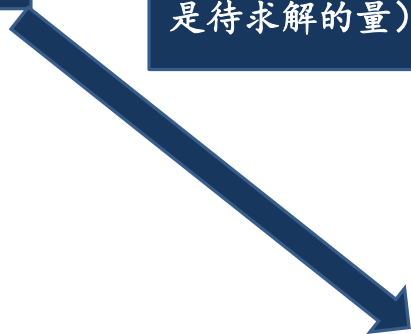
视觉里程计 (相机定位, 单目RGB相机情景下, 传统方法中深度信息与相机位姿都是待求解的量)



非线性优化



建图 (建立
场景地图)



回环检测



实例1

- ◆ CNN-SLAM: Real-time dense monocular SLAM with learned depth prediction
- ◆ 深度预测：使用CNN，使用预训练模型ResNet-50
- ◆ 深度获取加入语义分割
- ◆ 相机位姿预测：传统方法，关键帧
- ◆ 语义分割：使用CNN，与深度使用相同模型

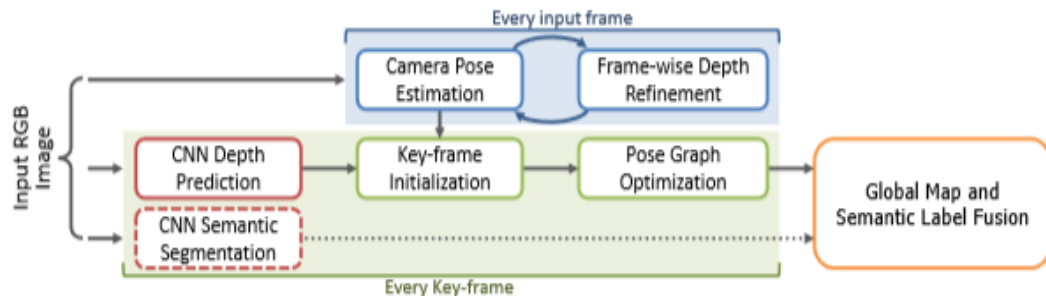


Figure 2. CNN-SLAM Overview.

实例2

◆ DeepVO: Towards End-to-End Visual Odometry with Deep Recurrent Convolutional Neural Networks

◆ 端到端的视觉里程计

- 输入：图像序列
- 输出：相机的pose

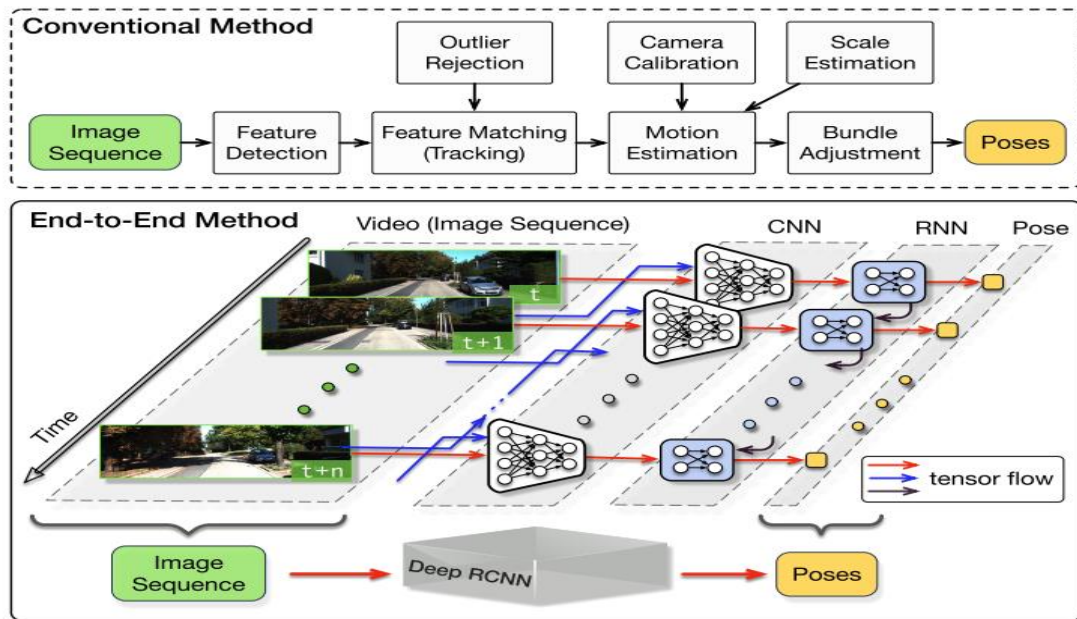


Fig. 1. Architectures of the conventional feature based monocular VO and the proposed end-to-end method. In the proposed method, RCNN takes a sequence of RGB images (video) as input and learns features by CNN for RNN based sequential modelling to estimate poses. Camera image credit: KITTI dataset [3].

◆ 网络结构：CNN+RNN

CNN提取特征，LSTM回归预测

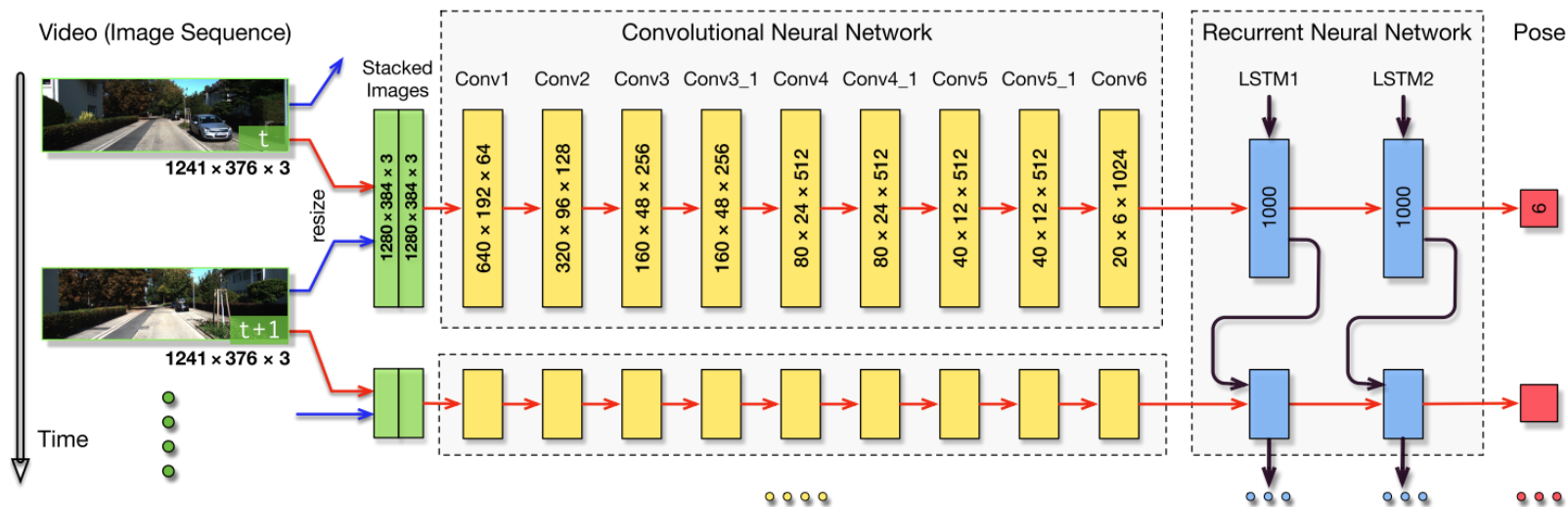


Fig. 2. Architecture of the proposed RCNN based monocular VO system. The dimensions of the tensors shown here are given as an example based on the image size of the KITTI dataset. The CNN ones should vary according to the size of the input image. Camera image credit: KITTI dataset.

LSTM序列的作用

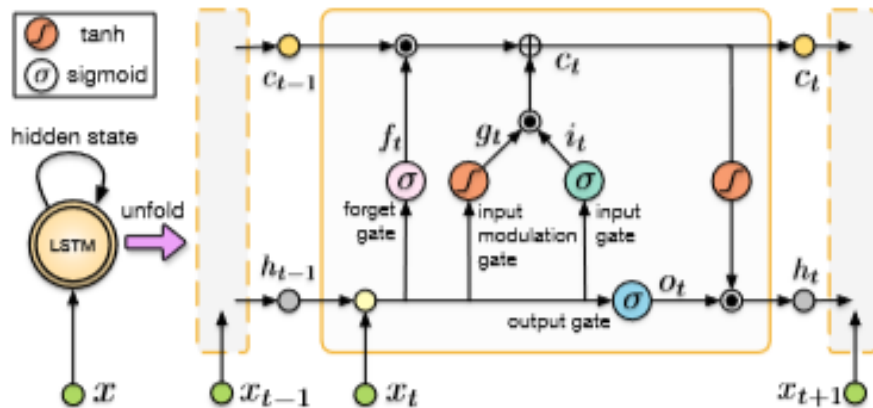
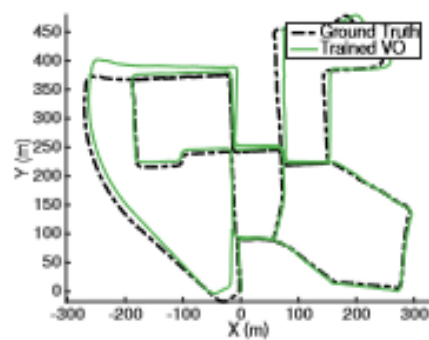
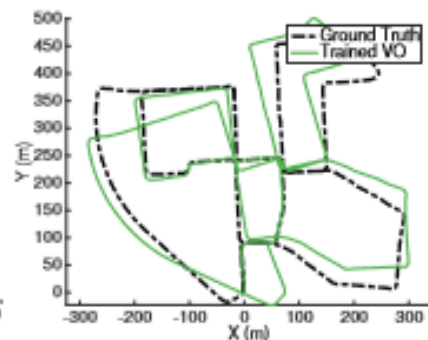


Fig. 3. Folded and unfolded LSTMs and internal structure of its unit. $\odot$ and $\oplus$ denote element-wise product and addition of two vectors, respectively.

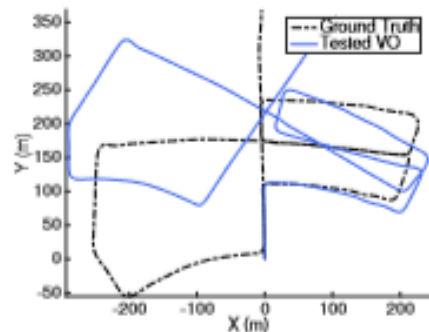
$$\begin{aligned}i_k &= \sigma(\mathbf{W}_{xi}\mathbf{x}_k + \mathbf{W}_{hi}\mathbf{h}_{k-1} + \mathbf{b}_i) \\f_k &= \sigma(\mathbf{W}_{xf}\mathbf{x}_k + \mathbf{W}_{hf}\mathbf{h}_{k-1} + \mathbf{b}_f) \\g_k &= \tanh(\mathbf{W}_{xg}\mathbf{x}_k + \mathbf{W}_{hg}\mathbf{h}_{k-1} + \mathbf{b}_g) \\c_k &= f_k \odot c_{k-1} + i_k \odot g_k \\o_k &= \sigma(\mathbf{W}_{xo}\mathbf{x}_k + \mathbf{W}_{ho}\mathbf{h}_{k-1} + \mathbf{b}_o) \\h_k &= o_k \odot \tanh(c_k)\end{aligned}$$



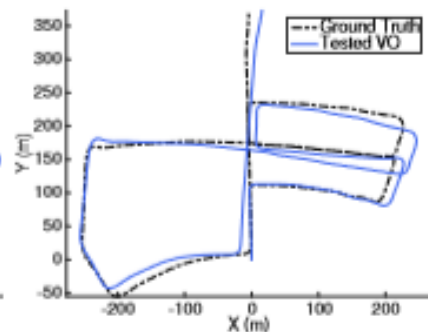
(c) Trained VO: Overfitting.



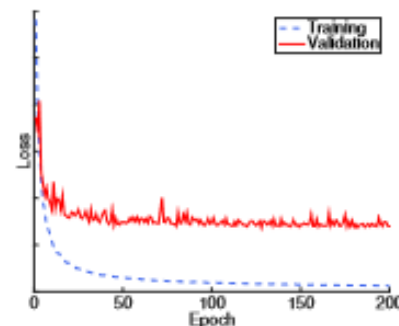
(d) Trained VO: Good Fit.



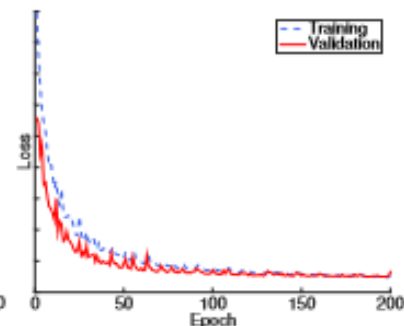
(e) Tested VO: Overfitting.



(f) Tested VO: Good Fit.

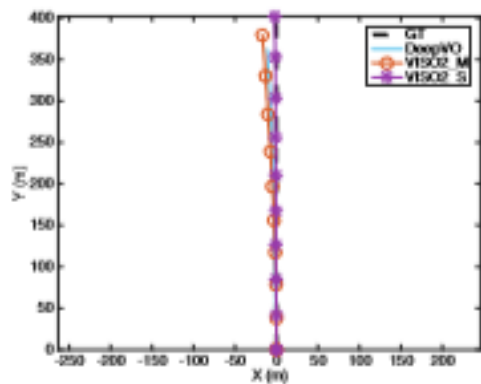


(a) Losses: Overfitting.

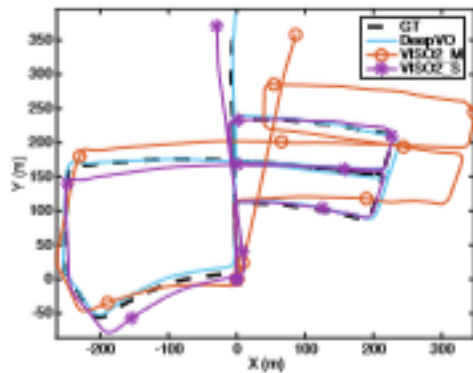


(b) Losses: Good Fit.

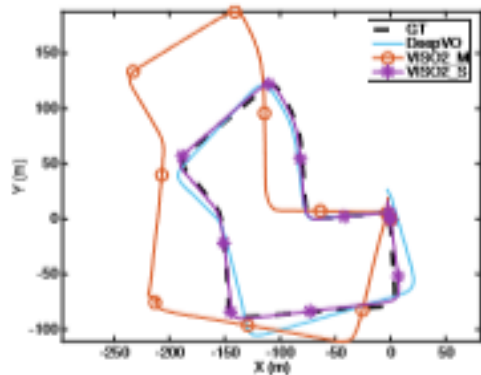
Fig. 4. Training losses and VO results of two models. Figures in the left and right columns are about the over-fitted and well-fitted models, respectively. (a)-(b) Training and validation losses. (c)-(d) Estimated VO on training data (Sequence 00). (e)-(f) Estimated VO on testing data (Sequence 05).



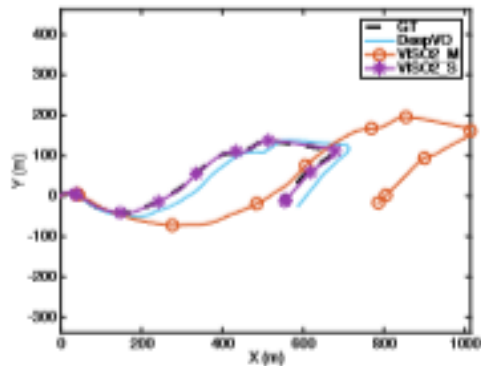
(a) Sequence 04.



(b) Sequence 05.



(c) Sequence 07.



(d) Sequence 10.

Fig. 6. Trajectories of VO testing results on Sequence 04, 05, 07 and 10. The DeepVO model used is trained on Sequence 00, 02, 08 and 09.

2. 从单幅图像重建

常见的从单幅图像重建算法包括：

- ◆ 从单幅图像生成三维点云
- ◆ 单幅图像恢复深度
- ◆ 从单幅图像生成网格

(1) 从单幅图像生成三维点云

- ◆ 输入：一幅图像
- ◆ 输出：三维点云坐标

从单幅图像生成三维点云实例1

- ◆ Fan H , Hao S , Guibas L . A Point Set Generation Network for 3D Object Reconstruction from a Single Image[C]// 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, 2017.
- ◆ CVPR 2017年

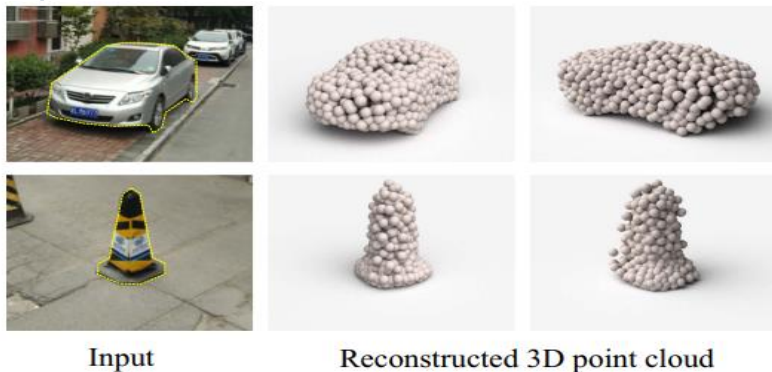


Figure 1. A 3D point cloud of the **complete** object can be reconstructed from a single image. Each point is visualized as a small sphere. The reconstruction is viewed at two viewpoints (0° and 90° along azimuth). A segmentation mask is used to indicate the scope of the object in the image.

A Point Set Generation Network for 3D Object Reconstruction from a Single Image

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- ◆ 深度学习方法，采用一个条件采样器，可以根据一张输入图像预测出多个合理的三维点云，进一步选择可信点云
- ◆ 例如，传统的方法就有利用一定特征作约束的方法：
 - 环境照明条件作假设：
 - ✓ B. K. Horn. **Obtaining shape from shading information**. In Shape from shading, pages 123–171. MIT press, 1989.
 - ✓ Aloimonos. **Shape from texture**. Biological cybernetics, 58(5):345–360, 1988.

从单幅图像生成三维点云

- ◆ 编码阶段：编码器由卷积层和ReLU层组成，将输入图像I和随机向量r映射到嵌入空间
- ◆ 预测阶段：预测器通过全连接网络，输出N行3列的矩阵M，其中每行是一个点的坐标，共N个点

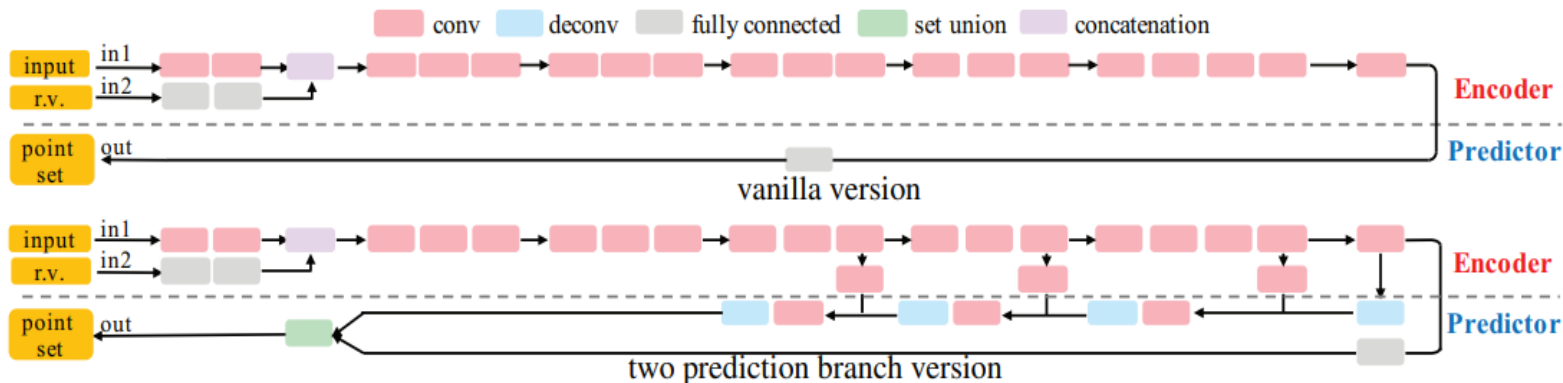


Figure 2. PointOutNet structure

从单幅图像生成三维点云

- ◆ 双预测分支版本，可以更好适应自然物体中常见表面，不同于全连接每个点独立预测的方式
- ◆ 两个平行预测器分支FC分支：同简单版本一样预测个点，灵活度高
- ◆ 更好地描述复杂结构deconv分支：得到一张 $H*W*3$ 通道的图像，其中每个像素的3个通道值是一个点的坐标
- ◆ 采用了跳跃连接

- ◆ 结果为无序点集（点云）：

个数：N=1024

- ◆ 监督学习

We seek for a distance d between subsets in $\mathbb{R}^3$, so that the loss function $L(\{S_i^{pred}\}, \{S_i^{gt}\})$ takes the form

$$L(\{S_i^{pred}\}, \{S_i^{gt}\}) = \sum_i d(S_i^{pred}, S_i^{gt}),$$

从单幅图像生成三维点云

- ◆ 损失计算方法:
- ◆ 点集间的距离度量

$$L(\{S_i^{pred}\}, \{S_i^{gt}\}) = \sum_i d(S_i^{pred}, S_i^{gt}),$$

- ◆ Chamfer distance (CD) 距离:
$$d_{CD}(S_1, S_2) = \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2^2 + \sum_{y \in S_2} \min_{x \in S_1} \|x - y\|_2^2$$
- ◆ Earth Mover's distance (EMD)
$$d_{EMD}(S_1, S_2) = \min_{\phi: S_1 \rightarrow S_2} \sum_{x \in S_1} \|x - \phi(x)\|_2$$
- ◆ 存在的问题:
 - 神经网络预测时会有不确定性

实验结果

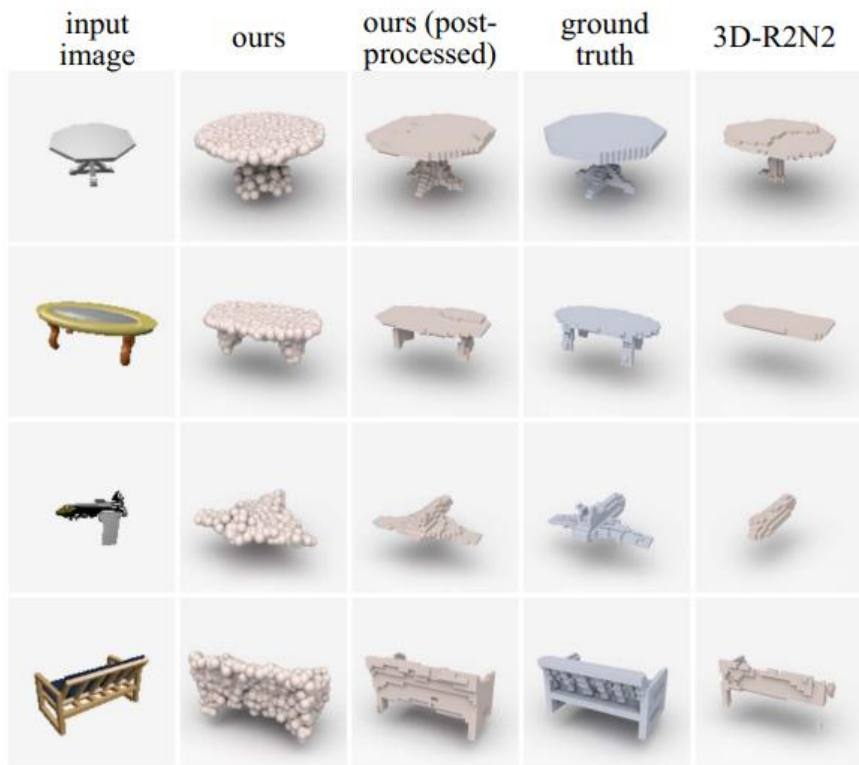


Figure 5. Visual comparison to 3D-R2N2. Our method better preserves thin structures of the objects.

从单幅图像重建点云-实例2

Pixel2point: 3D Object Reconstruction From a Single Image Using CNN and Initial Sphere

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- ◆ set the number of the points generated from the CNN to $N = 2048$
- ◆ this number of points is sufficient to cover the whole surface of the object and preserves the major structures.

从单幅图像重建

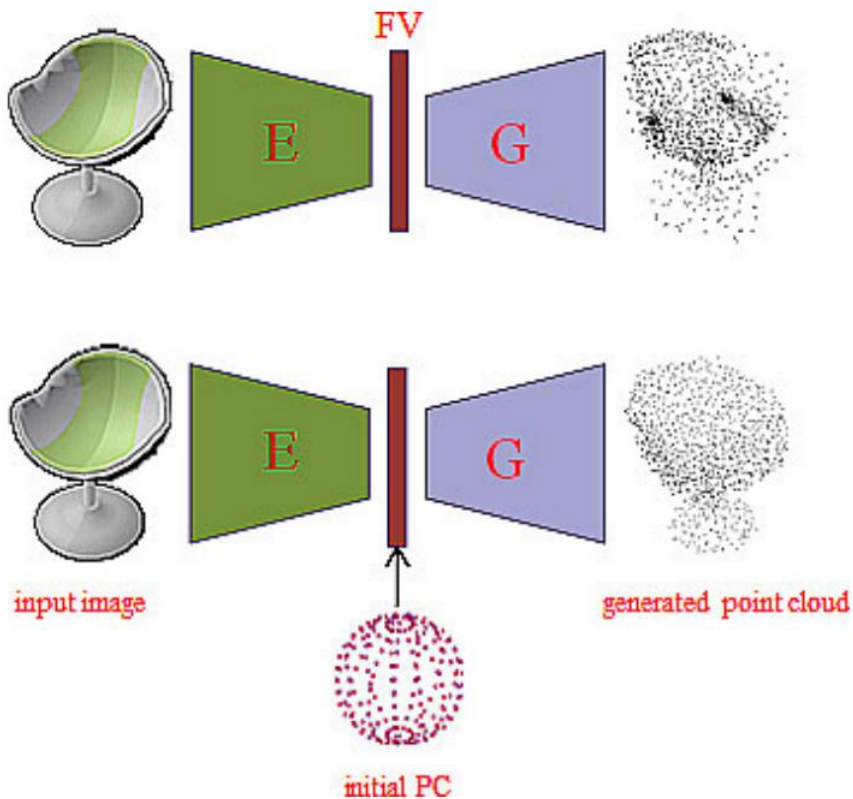


FIGURE 1. A general sketch of the proposed CNN model with different setups. Top: the proposed CNN without an initial point cloud. Bottom: the proposed CNN with the initial point cloud. E: Encoder, G: Generator, PC: Point Cloud, FV: Feature Vector.

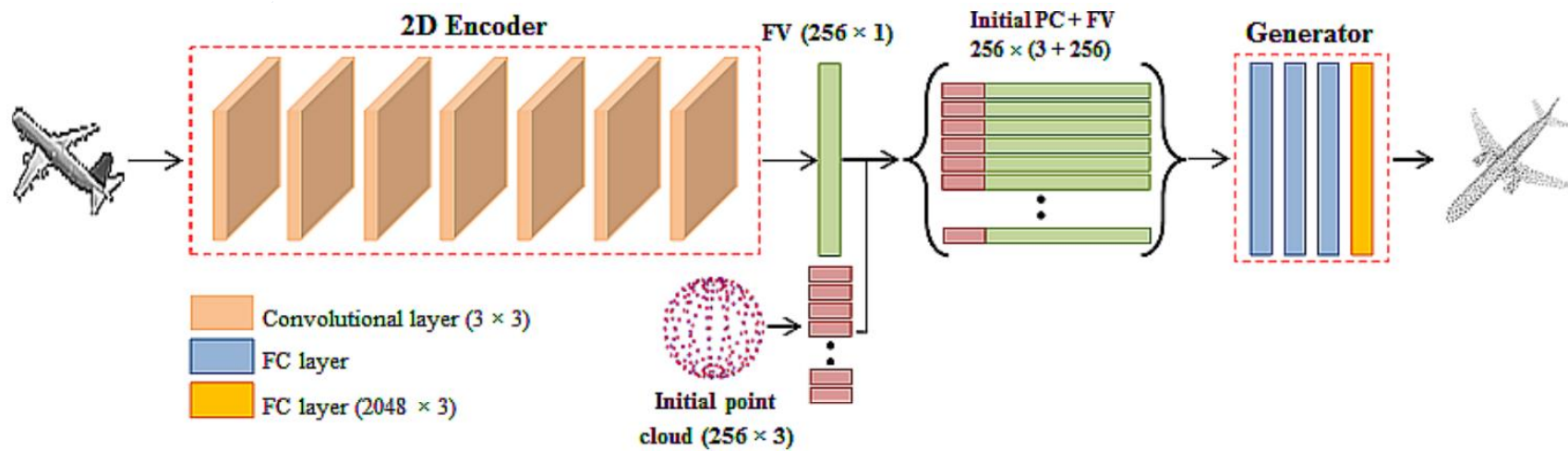


FIGURE 2. The proposed CNN Architecture with the initial Point Cloud.

损失函数

So, the required loss function L between two 3D shapes, $S^{pred}, S^{gt} \subseteq \mathbb{R}^3$, is defined as:

$$L(\{S^{pred}\}, \{S^{gt}\}) = \sum d(S^{pred}, S^{gt}) \quad (2)$$

Chamfer Distance (CD). The Chamfer Distance between $S_1, S_2 \subseteq \mathbb{R}^3$ is defined as:

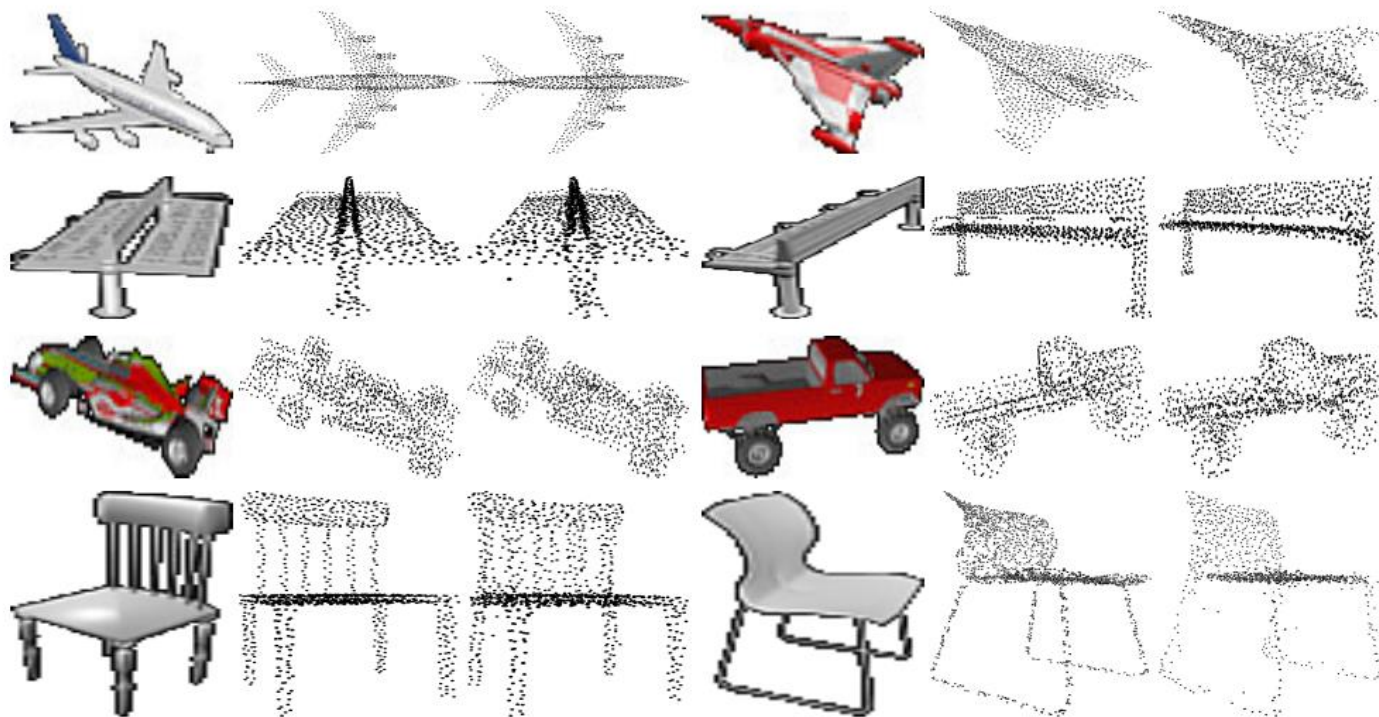
$$d_{CD} = \sum_{x \in S_1} \min_{y \in S_2} \|x - y\|_2^2 + \sum_{y \in S_2} \min_{x \in S_1} \|x - y\|_2^2 \quad (3)$$

Earth Mover's Distance (EMD). The EMD between $S_1, S_2 \subseteq \mathbb{R}^3$ is defined as:

$$d_{EMD} = \min_{\phi: S_1 \rightarrow S_2} \sum_{x \in S_1} \|x - \phi(x)\|_2 \quad (4)$$

where $\phi : S_1 \rightarrow S_2$ is a bijection and the size of S_1 and S_2 is equal, $s = |S_1| = |S_2|$.

结果





(2) 单幅图像恢复深度

实例

Depth Map Prediction from a Single Image using a Multi-Scale Deep Network

- ◆ Input: 单张RGB图片
- ◆ Output: 该图像对应的深度图

网络结构

- ◆ 第一个网络得到全局深度图的预测
- ◆ 第二个网络根据局部信息对全局深度图进行细节调整，即将像素之间的深度关系进行考虑

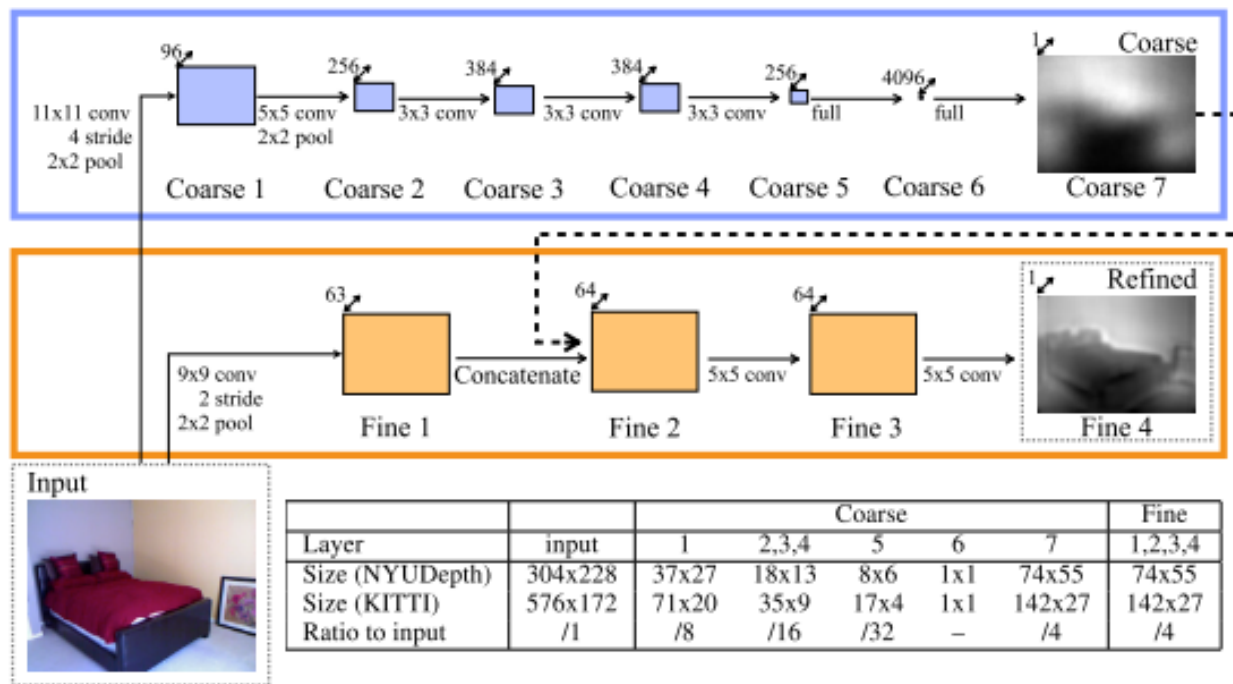
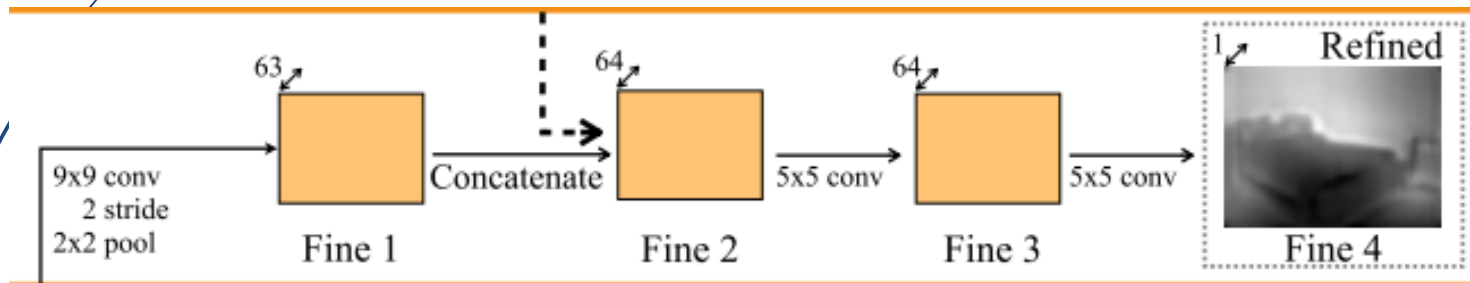


Figure 1: Model architecture.

Local Fine-Scale Network

- ◆ 边界接收到粗预测，结合物体和边缘局部细节
- ◆ 只包含卷积层
- ◆ 粗略网络的输出作为低级特征图输入



粗细网络的结合

- ◆ 两个网络分开训练，先使用ground truth训练粗网络，得到输出，再将结果用来训练第二个网络
- ◆ 两个网络各自采用反向传播

尺度不变误差

i, j 像素的相对误差 像素间调整误差

$$D(y, y^*) = \frac{1}{2n^2} \sum_{i,j} ((\log y_i - \log y_j) - (\log y_i^* - \log y_j^*))^2 \quad (2)$$

$$= \frac{1}{n} \sum_i d_i^2 - \frac{1}{n^2} \sum_{i,j} d_i d_j = \frac{1}{n} \sum_i d_i^2 - \frac{1}{n^2} \left(\sum_i d_i \right)^2 \quad (3)$$

深度的距离误差

each pair of pixels in the prediction must differ in depth by an amount similar to that of the corresponding pair in the ground truth

Depth Map Prediction: loss

- ◆ $\lambda \in [0, 1]$,
- ◆ $\lambda=0$ 时, L 为单纯的距离误差
- ◆ $\lambda=1$ 时, L 为尺度不变误差
- ◆ 论文中 $\lambda=0.5$

$$L(y, y^*) = \frac{1}{n} \sum_i d_i^2 - \frac{\lambda}{n^2} \left(\sum_i d_i \right)^2 \quad (4)$$

实验结果

◆ 从左至右依次为：

- 输入
- 粗尺度输出
- 细尺度输出
- Ground truth



深度估计的其它前沿方法

2024

-    Ruizhuo Xu , Ke Wang, Chao Deng, Mei Wang , Xi Chen, Wenhui Huang, Junlan Feng, Weihong Deng 
:
Depth Map Denoising Network and Lightweight Fusion Network for Enhanced 3D Face Recognition. Pattern Recognit. 145: 109936 (2024)

2023

-    Il-Youp Kwak , Sungsu Kwag , Junhee Lee , Youngbae Jeon, Jeong-Hwan Hwang, Hyoukryeol Choi, Jong-Hoon Yang, Soyul Han, Jun Ho Huh , Choong-Hoon Lee , Ji Won Yoon 
Voice Spoofing Detection Through Residual Network, Max Feature Map, and Depthwise Separable Convolution. IEEE Access 11: 49140-49152 (2023)
-    Fengqin Wang, Zhiying Wang , Qiuwen Zhang 
CNN-LNN Based Fast CU Partitioning Decision for VVC 3D Video Depth Map Intra Coding. IEEE Access 11: 87420-87429 (2023)
-    Ting Zhang , Bo Tian, Yujue Wang, Dongyan Liu, Yunxuan Zhou , Daphne van der Wal 
Mapping depth-integrated microphytobenthic biomass on an estuarine tidal flat using Sentinel satellite data. Int. J. Appl. Earth Obs. Geoinformation 122: 103417 (2023)
-    Yuyang Wang, Jingyu Yang, Huanjing Yue:
Depth map continuous super-resolution with local implicit guidance function. Displays 78: 102418 (2023)
-    Maxim V. Bobyr , Alexander Arkhipov , Sergey G. Emelyanov , Natalya A. Milostnaya:
A method for creating a depth map based on a three-level fuzzy model. Eng. Appl. Artif. Intell. 117(Part): 105629 (2023)
-    Xiaoming Ding, Liang Hu, Shubo Zhou, Xiaocheng Wang, Yupeng Li, Tingting Han, [Dunqiang Lu](#), Guowei Che:
Snapshot depth-spectral imaging based on image mapping and light field. EURASIP J. Adv. Signal Dunqiang Lu

(3) 从单幅图像生成网格模型—Pixel2Mesh

- ◆ Nanyang Wang, Yinda Zhang, Zhuwen Li, Yanwei Fu, Wei Liu, Yu-Gang Jiang: Pixel2Mesh: Generating 3D Mesh Models from Single RGB Images. ECCV (11) 2018: 55–71

Pixel2Mesh: Generating 3D Mesh Models from Single RGB Images

Nanyang Wang¹, Yinda Zhang², Zhuwen Li³, Yanwei Fu⁴, Wei Liu⁵,
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² Princeton University, Princeton, USA
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³ Intel Labs, Santa Clara, USA
lzhuwen@gmail.com

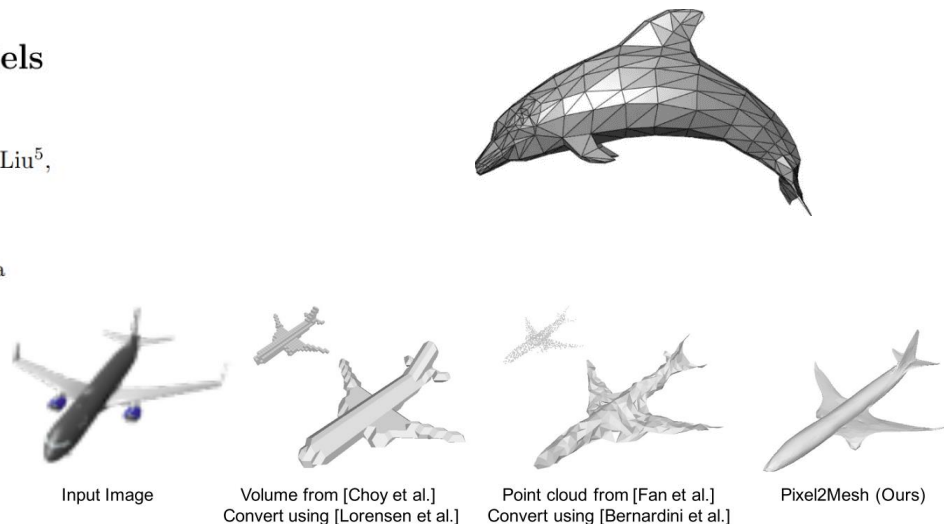


Fig. 1. Given a single color image and an initial mesh, our method can produce a high-quality mesh that contains details from the example.

思想

1. 首先给定一张输入图像：Input Image。
2. 为任意的输入图像都初始化一个固定大小的椭球体（三轴半径分别为0.2、0.2、0.8m）作为其初始三维形状：Ellipsoid Mesh。

网络结构

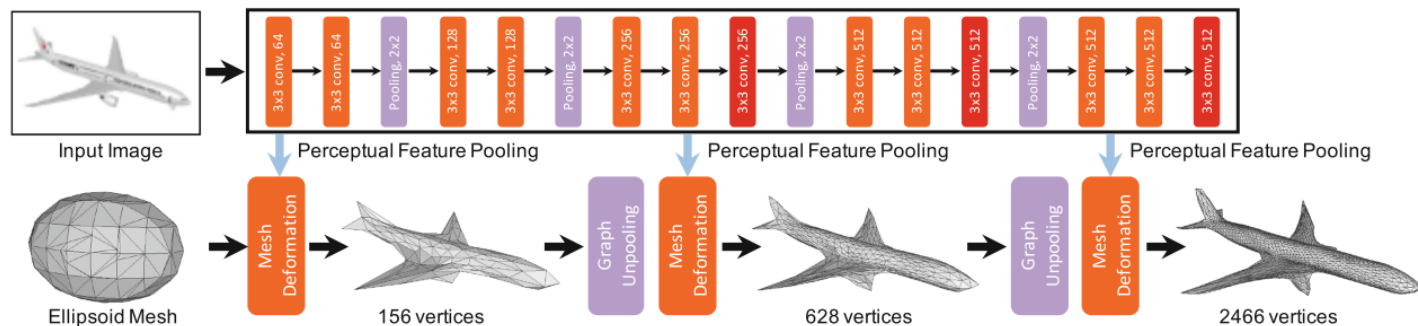
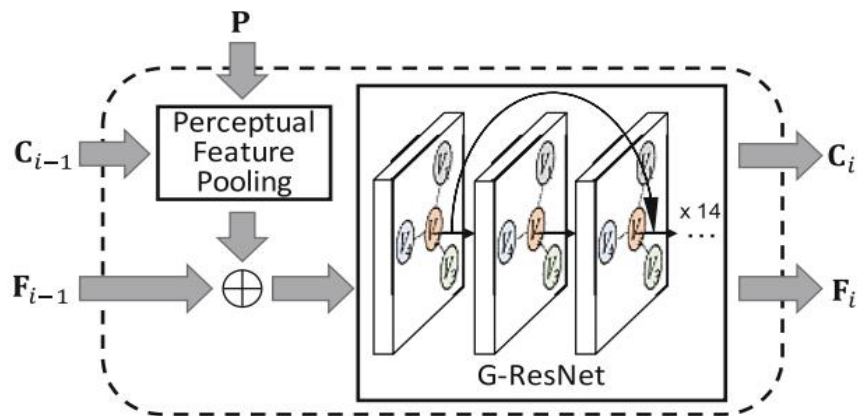


Fig. 2. The cascaded mesh deformation network. Our full model contains three mesh deformation blocks in a row. Each block increases mesh resolution and estimates vertex locations, which are then used to extract perceptual image features from the 2D CNN for the next block.

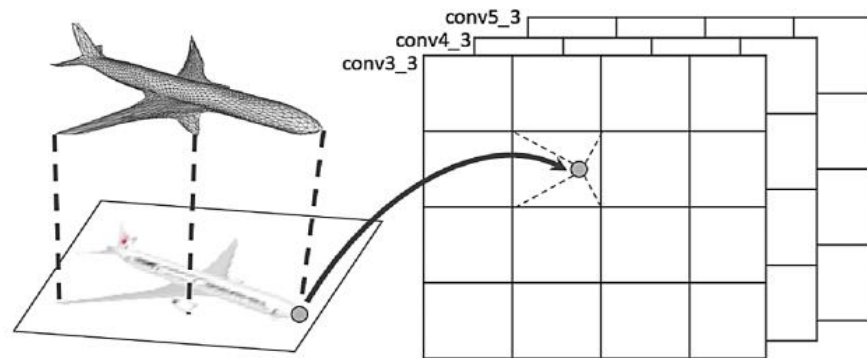
整个网络可以分成上下两个部分：图像特征提取网络和级联网格变形网络。

- ◆ 上面部分负责用全卷积神经网络（CNN）提取输入图像的特征。
- ◆ 下面部分负责用图卷积神经网络（GCN）来提取三维mesh特征，并不断地对三维mesh进行形变，逐步将椭球网格变形为所需的三维模型，目标是得到最终的飞机模型。
- ◆ 图卷积神经网络提取特征：
 - GCN其实就是针数据结构中一种形式图的结构做特征提取。GCN在表示3D结构上具有优势
 - 3D mesh是由顶点、边和面来描述三维对象的，正好对应于图卷积神经网络 $G = (V, E, F)$ ，分别为顶点Vertex、边Edge和特征向量Feature。

Mesh变形模块



(a) Mesh Deformation Block



(b) Perceptual Feature Pooling

Fig. 3. (a) The vertex locations C_i are used to extract image features, which are then combined with vertex features F_i and fed into G-ResNet. $\oplus$ means a concatenation of the features. (b) The 3D vertices are projected to the image plane using camera intrinsics, and perceptual feature is pooled from the 2D-CNN layers using bilinear interpolation.

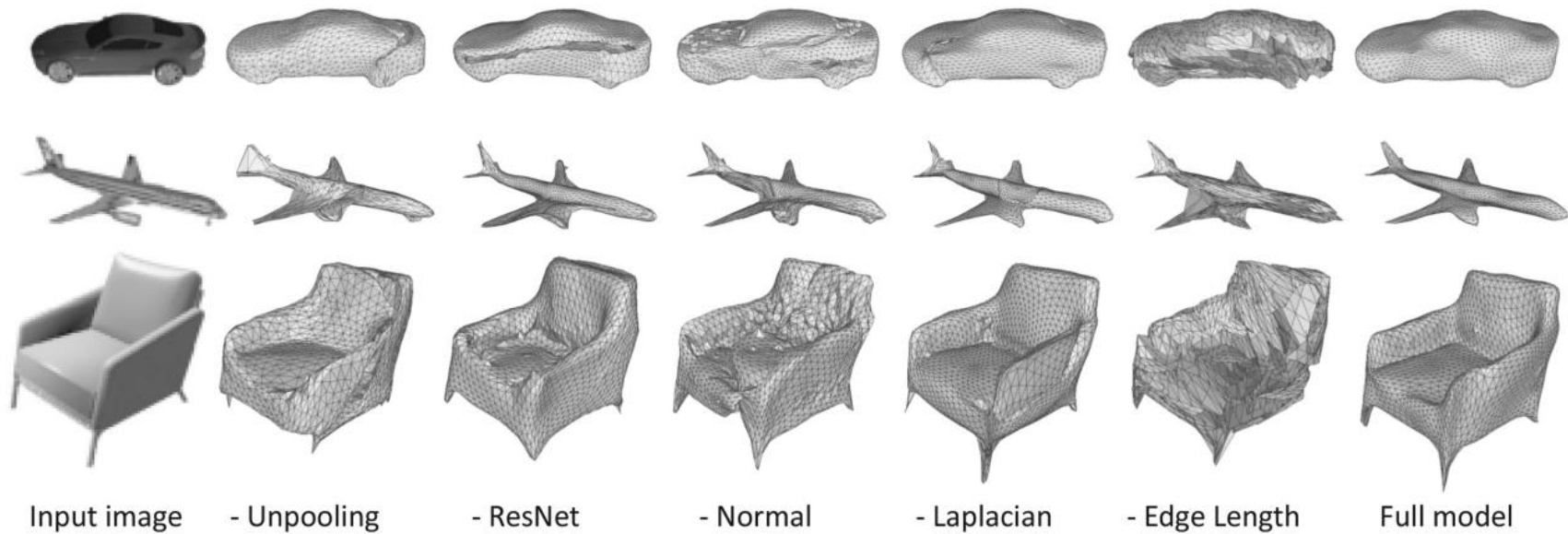


Fig. 5. Qualitative results for ablation study. This figure truly reflects the contribution of each components especially for the regularization ones.

其它文献

2023

-    Chao Wen , Yinda Zhang, Chenjie Cao, Zhuwen Li , Xiangyang Xue , Yanwei Fu :
Pixel2Mesh++: 3D Mesh Generation and Refinement From Multi-View Images. IEEE Trans. Pattern Anal. Mach. Intell. 45(2): 2166-2180 (2023)
-    Rongshan Chen, Xiang Yin, Yuancheng Yang, Chao Tong:
Multi-view Pixel2Mesh++: 3D reconstruction via Pixel2Mesh with more images. Vis. Comput. 39(10): 5153-5166 (2023)

2022

-    Chao Wen, Yinda Zhang, Chenjie Cao, Zhuwen Li, Xiangyang Xue, Yanwei Fu:
Pixel2Mesh++: 3D Mesh Generation and Refinement from Multi-View Images. CoRR abs/2204.09866 (2022)

2021

-    Nanyang Wang, Yinda Zhang, Zhuwen Li , Yanwei Fu , Hang Yu, Wei Liu , Xiangyang Xue , Yu-Gang Jiang :
Pixel2Mesh: 3D Mesh Model Generation via Image Guided Deformation. IEEE Trans. Pattern Anal. Mach. Intell. 43(10): 3600-3613 (2021)

2019

-    Chao Wen, Yinda Zhang, Zhuwen Li, Yanwei Fu:
Pixel2Mesh++: Multi-View 3D Mesh Generation via Deformation. ICCV 2019: 1042-1051
-    Chao Wen, Yinda Zhang, Zhuwen Li, Yanwei Fu:
Pixel2Mesh++: Multi-View 3D Mesh Generation via Deformation. CoRR abs/1908.01491 (2019)

2018

-    Nanyang Wang, Yinda Zhang, Zhuwen Li, Yanwei Fu, Wei Liu , Yu-Gang Jiang:
Pixel2Mesh: Generating 3D Mesh Models from Single RGB Images. ECCV (11) 2018: 55-71

3. 基于NerF的技术

- ◆ NeRF (Neural Radiance Fields神经辐射场) :
- ◆ 利用隐式表达
- ◆ 最早在2020年ECCV会议上获得最佳论文, 2D的images作为监督, 提供poses, 可表示复杂的三维场景。

2022



Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, Ren Ng: **NeRF: representing scenes as neural radiance fields for view synthesis**. Commun. ACM 65(1): 99-106 (2022)

2020



Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, Ren Ng: **NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis**. ECCV (1) 2020: 405-421



Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, Ren Ng: **NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis**. CoRR abs/2003.08934 (2020)

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis

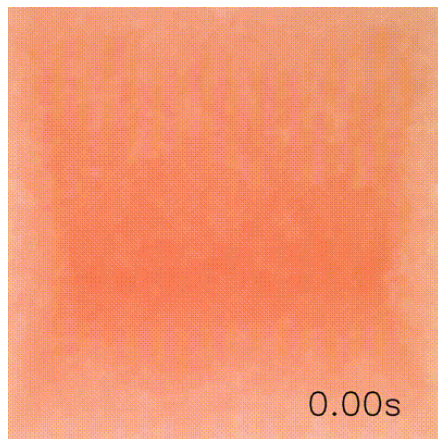
Ben Mildenhall^{1*} Pratul P. Srinivasan^{1*} Matthew Tancik^{1*}
Jonathan T. Barron² Ravi Ramamoorthi³ Ren Ng¹

¹UC Berkeley ²Google Research ³UC San Diego

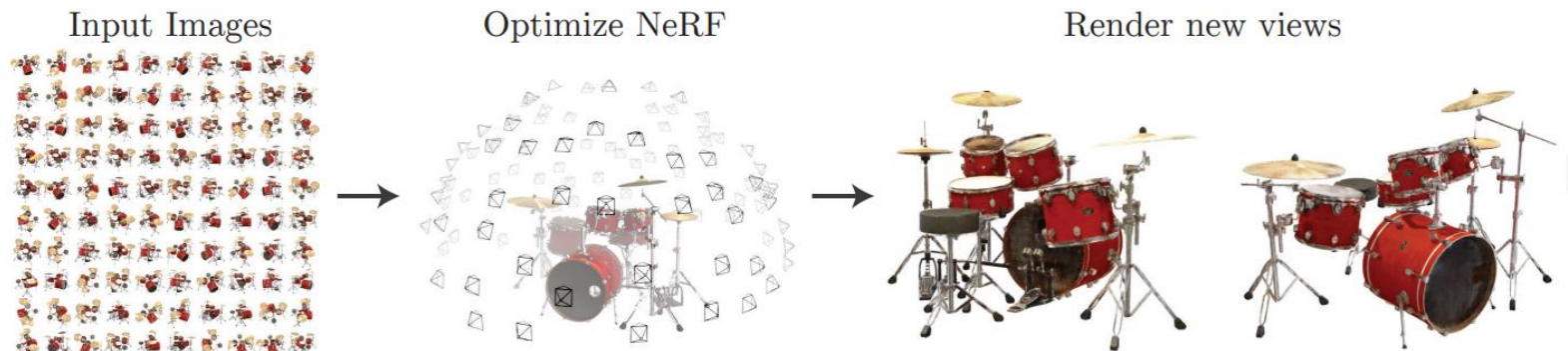
Abstract. We present a method that achieves state-of-the-art results for synthesizing novel views of complex scenes by optimizing an underlying continuous volumetric scene function using a sparse set of input views. Our algorithm represents a scene using a fully-connected (non-

设计思想

- ◆ 采用新的视点合成思想
- ◆ 计算每个采集角度的相机位姿Pose，将采集的图像序列以及pose输入，合成新视点的结果



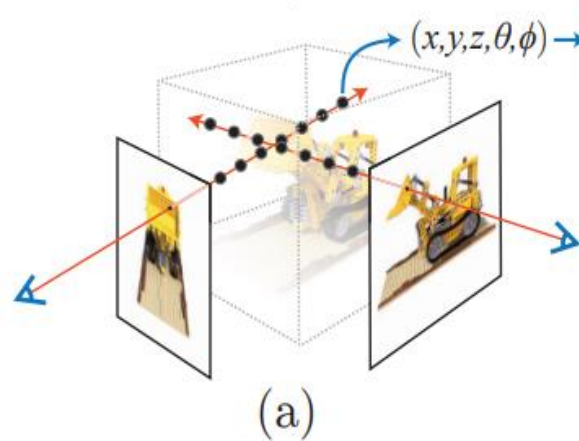
- ◆ 从一组输入图像中优化场景的连续5D神经辐射场表示、
- ◆ 以从任何视点渲染场景
- ◆ 任何连续位置的体积密度和视图相关颜色
- ◆ 用体积渲染的技术来沿着光线累积表示场景采样
- ◆ 实例：Drums场景，100个输入视图



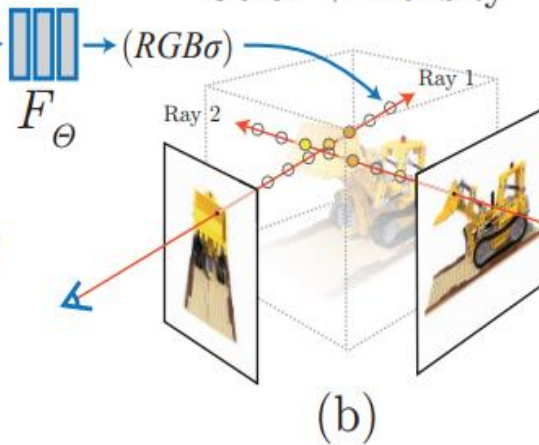
◆ 连续场景表示为5D向量值函数：

- 输入是3D位置 $x = (x, y, z)$ 和2D视点方向 (θ, ϕ)
- 输出是辐射颜色 $c = (r, g, b)$ 和体密度 σ

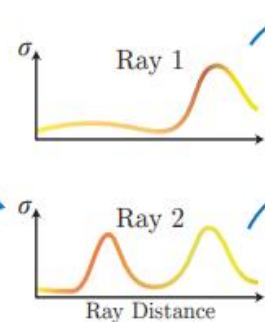
5D Input
Position + Direction



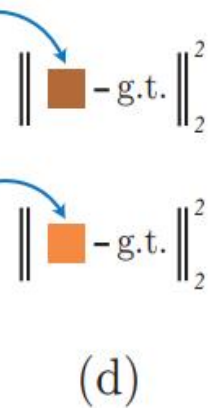
Output
Color + Density



Volume
Rendering



Rendering
Loss



- ◆ 网络输入五个自由度：
- ◆ 输出：颜色和密度（只与xyz有关）
- ◆ 颜色与（xyz和视点方向有关）
- ◆ 用实际颜色作监督

该算法存在问题

- (1) 速度慢：训练和渲染时间长
- (2) 针对静态场景
- (3) 训练结果不具有通用性，泛化能力差

出现的改进版本：

Mip-NeRF、NeRF++、Block-NeRF、Urban-NeRF、CityNeRF等

NerF的现有前沿工作

2023

-    Hyunjin Kim , Daekyeong Lee, Suyoung Kang, Pyojin Kim 
Complex-Motion NeRF: Joint Reconstruction and Pose Optimization With Motion and Depth Priors. IEEE Access 11: 97425-97434 (2023)
-    Guo-Wei Yang, Zheng-Ning Liu, Dong-Yang Li, Hao-Yang Peng:
JNeRF: An efficient heterogeneous NeRF model zoo based on Jittor. Comput. Vis. Media 9(2): 401-404 (2023)
-    Junyu Zhu, Hao Zhu, Qi Zhang , Fang Zhu, Zhan Ma, Xun Cao:
Pyramid NeRF: Frequency Guided Fast Radiance Field Optimization. Int. J. Comput. Vis. 131(10): 2649-2664 (2023)
-    Zhenbiao Gai, Zhenyang Liu, Min Tan, Jiajun Ding, Jun Yu, Mingzhao Tong, Junqing Yuan:
EGRA-NeRF: Edge-Guided Ray Allocation for Neural Radiance Fields. Image Vis. Comput. 134: 104670 (2023)
-    Clément Jambon , Bernhard Kerbl , Georgios Kopanas , Stavros Diolatzis , Thomas Leimkühler , George Drettakis 
NeRFshop: Interactive Editing of Neural Radiance Fields Proc. ACM Comput. Graph. Interact. Tech. 6(1): 1:1-1:21 (2023)
-    Zackary P. T. Sin , Peter H. F. Ng , Hong Va Leong 
NeRFahedron: A Primitive for Animatable Neural Rendering with Interactive Speed. Proc. ACM Comput. Graph. Interact. Tech. 6(1): 2:1-2:20 (2023)
-    Simon Klenk , Lukas Koestler, Davide Scaramuzza , Daniel Cremers 
E-NeRF: Neural Radiance Fields From a Moving Event Camera IEEE Robotics Autom. Lett. 8(3): 1587-1594 (2023)
-    Jiankai Sun , Yan Xu , Mingyu Ding , Hongwei Yi, Chen Wang , Jingdong Wang, Liangjun Zhang , Mac Schwager 
NeRF-Loc: Transformer-Based Object Localization Within Neural Radiance Fields. IEEE Robotics

4. 3D Gaussian Splatting的重建技术

(1) 3D Gaussian Splatting 在2023年ACM Trans. Graph中发表:

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, George Drettakis: 3D Gaussian Splatting for Real-Time Radiance Field Rendering. ACM Trans. Graph. 42(4): 139:1-139:14 (2023)

研究的出发点:

- ◆ 体素辐射场方法，具有一定优点，但高质量的视觉效果需要神经网络，训练和渲染成本高
- ◆ 本文引入了三个关键要素，改善实时的重建效果
- ◆ 能够在 1080p 分辨率下实现高质量的实时 (≥ 30 fps) 新视图合成。

◆ 3D Gaussian Splatting技术特点:

- 三维高斯作为场景表示法即使用SfM校准的像机, 并使用 SfM 过程中生成的稀疏点云初始化 3D 高斯集
- 优化三维高斯的属性--三维位置、不透明度

- ◆ 引入各向异性三维高斯作为辐射场的高质量、非结构化表示方法
- ◆ 针对GPU 的快速可微分渲染方法，进行各向异性拼接和快速后向传播，从而实现高质量的新视图合成

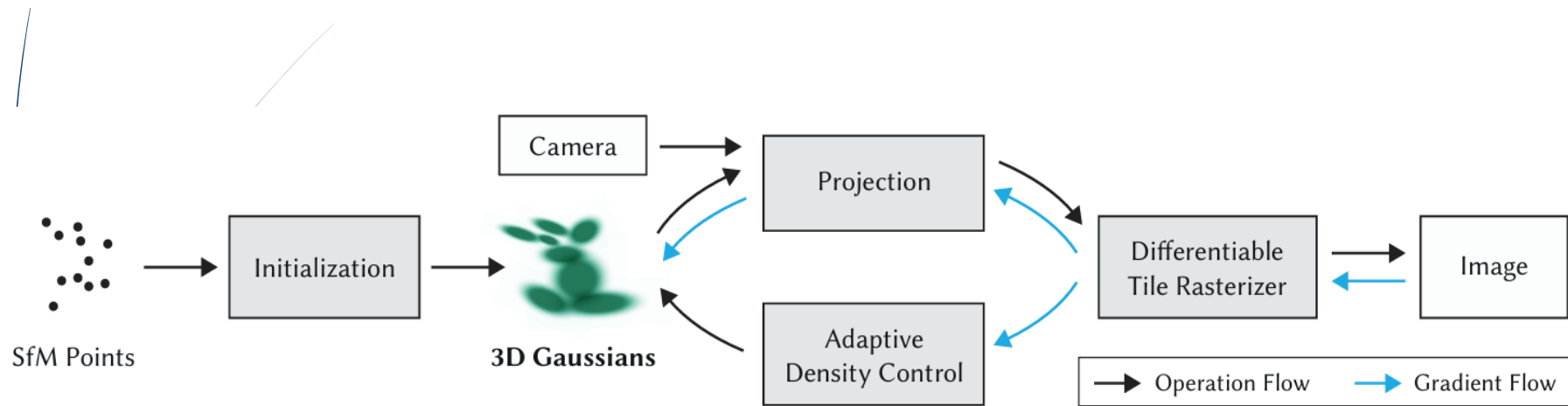


Fig. 2. Optimization starts with the sparse SfM point cloud and creates a set of 3D Gaussians. We then optimize and adaptively control the density of this set of Gaussians. During optimization we use our fast tile-based renderer, allowing competitive training times compared to SOTA fast radiance field methods. Once trained, our renderer allows real-time navigation for a wide variety of scenes.

可微分3D Gaussian Splatting

- ◆ 从无法向的稀疏 (SfM) 点集开始, 进行高质量的新视图合成
- ◆ 选择了三维高斯, 继承了可微分体积表示法的特性, 非结构化, 可以实现快速渲染
- ◆ 高斯是由以点 (平均值) μ 为中心的全三维协方差矩阵 Σ 定义

$$G(x) = e^{-\frac{1}{2}(x)^T \Sigma^{-1}(x)}$$

- ◆ 需要将三维高斯投影到二维空间, 以便进行渲染
- ◆ 直接优化协方差矩阵 Σ , 以获得代表辐射场的三维高斯

高斯自适应控制

- ◆ 自适应地控制单位体积内的高斯数量和密度，从初始稀疏的高斯集变为更密集的高斯集，从而以正确的参数更好地表现场景。
- ◆ 每迭代 100 次就会进行一次高斯致密化处理，并移除基本透明的高斯
- ◆ 对高斯的自适应控制需要填充空白区域。它不仅关注几何特征缺失的区域（"重建不足"），也关注高斯覆盖场景大面积的区域（通常对应 "重建过度"）
- ◆ 优化过程会出现对应的区域还没有得到很好的重构，而优化试图移动高斯来纠正这一现象

高斯自适应控制

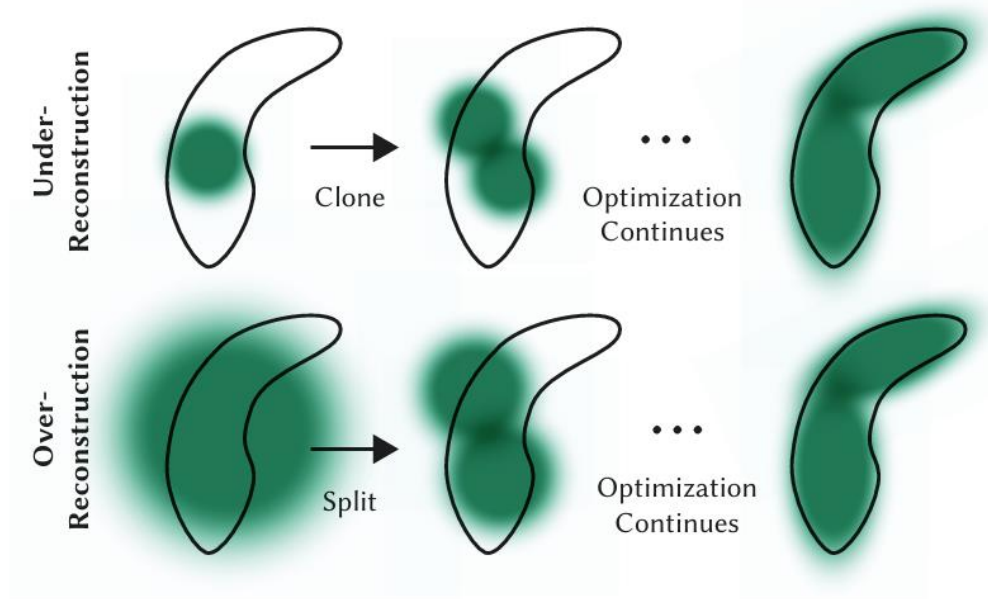


Fig. 4. Our adaptive Gaussian densification scheme. *Top row (under-reconstruction)*: When small-scale geometry (black outline) is insufficiently covered, we clone the respective Gaussian. *Bottom row (over-reconstruction)*: If small-scale geometry is represented by one large splat, we split it in two.

Ground Truth

Ours

Mip-NeRF360

InstantNGP

Plenoxels



5. 基于SDF的重建技术

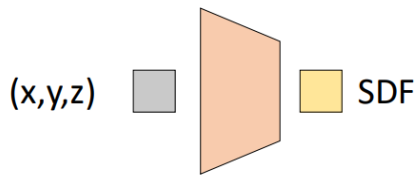
(1) DeepSDF

- ◆ 2019CVPR顶级会议文章:

Jeong Joon Park, Peter R. Florence, Julian Straub, Richard A. Newcombe, Steven Lovegrove:
DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation. CVPR 2019:
165-174

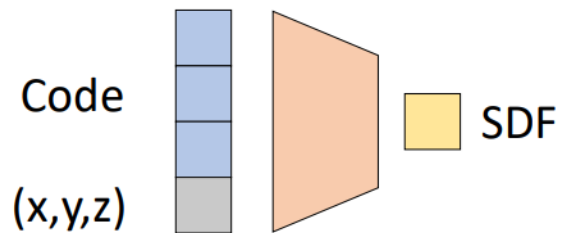
主要思想:

- ◆ 用SDF (Signed Distance Function) 函数来隐式表示三维物体
- ◆ 输入是三维空间点, 输出三维!



Single Shape DeepSDF

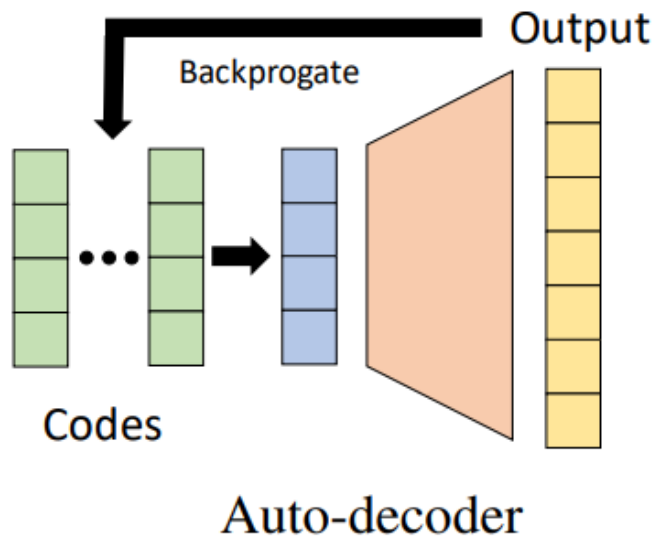
系统结构



Coded Shape DeepSDF

Encoder-decoder 结构

- ◆ 在训练阶段可以用编码-解码网络
- ◆ 实践应用时，仅仅使用解码器：固定解码器的参数，反向传播，优化中间的 latent code
- ◆ 采用 auto-decoder 的架构
- ◆ 训练阶段：A randomly initialized latent vector is assigned to each data point in the beginning of training, and the latent vectors are optimized along with the decoder weights through standard backpropagation.
- ◆ 预测阶段：During inference, decoder weights are fixed, and an optimal latent vector is estimated.



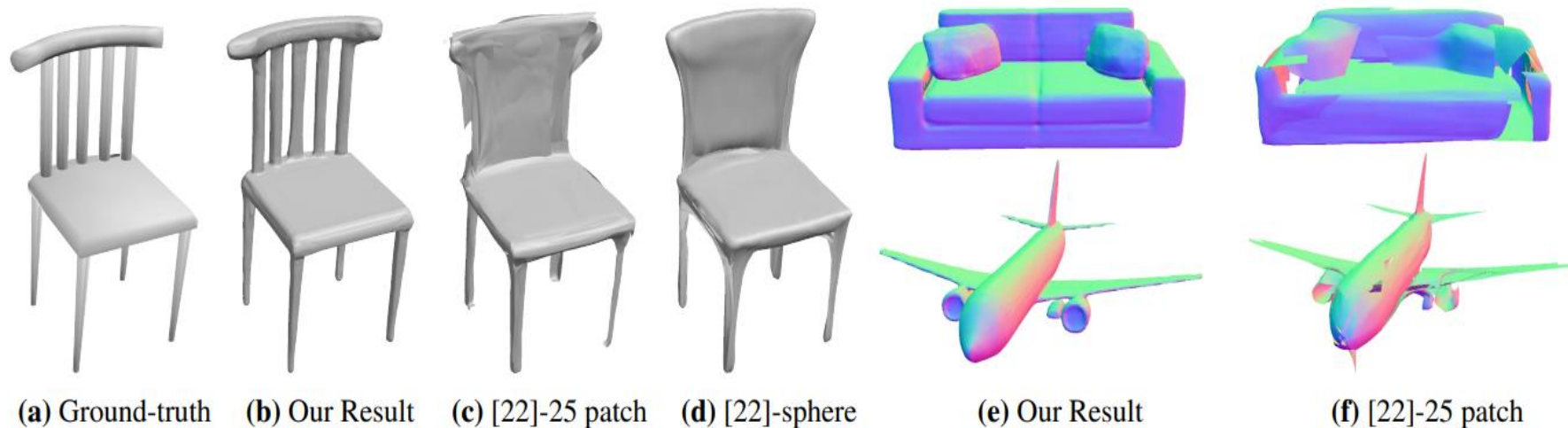


Figure 6: Reconstruction comparison between DeepSDF and AtlasNet [22] (with 25-plane and sphere parameterization) for test shapes. Note that AtlasNet fails to capture the fine details of the chair, and that (f) shows holes on the surface of sofa and the plane.

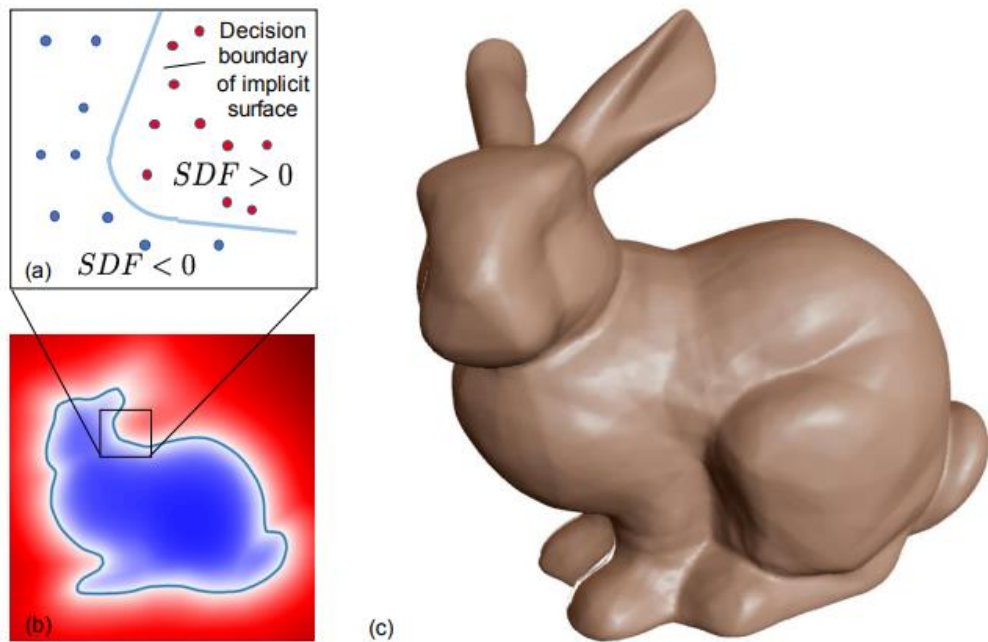


Figure 2: Our DeepSDF representation applied to the Stanford Bunny: (a) depiction of the underlying implicit surface $SDF = 0$ trained on sampled points inside $SDF < 0$ and outside $SDF > 0$ the surface, (b) 2D cross-section of the signed distance field, (c) rendered 3D surface recovered from $SDF = 0$. Note that (b) and (c) are recovered via DeepSDF.

(2) 隐神经模板Deep Implicit Templates 3D重建

- ◆ Zerong Zheng, Tao Yu, Qionghai Dai, Yebin Liu:Deep Implicit Templates for 3D Shape Representation. CVPR 2021: 1429-1439
- ◆ 2021, CVPR发表

Deep Implicit Templates for 3D Shape Representation

Zerong Zheng Tao Yu Qionghai Dai Yebin Liu

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Abstract

Deep implicit functions (DIFs), as a kind of 3D shape representation, are becoming more and more popular in the 3D vision community due to their compactness and strong representation power. However, unlike polygon mesh-based templates, it remains a challenge to reason dense correspondences or other semantic relationships across shapes

Unfortunately, the implicit nature of DIFs is also its Achilles' Heels: although DIFs are good at approximating individual shapes, they provide no information about the relationship between two different ones. One can easily establish vertice-to-vertice correspondences between two shapes when using mesh templates [21, 50, 14], but that is difficult in DIFs. The lack of semantic relationship in DIFs poses significant challenges for using DIFs in downstream appli-

- ◆ Deep implicit functions 是三维形状表示的一种方法，被应用于三维视觉中
- ◆ 文章提出3D 形状表示的隐模板法：Deep Implicit Templates，使 DIF 更具有可解释性

主要思想：

- 将 DIFs 表述为模板隐函数的条件变形。
- 提出Spatial Warping LSTM，将条件空间变换分解为多个仿射变换

- ◆ 对隐模板法公式化，它分解将条件有符号距离函数 F 转换为 $F=T \circ W$

$$\mathcal{F}(\mathbf{p}, \mathbf{c}) = \mathcal{T}(\mathcal{W}(\mathbf{p}, \mathbf{c}))$$

where $\mathcal{W} : \mathbb{R}^3 \times \mathcal{X} \mapsto \mathbb{R}^3$ maps the coordinate of $\mathbf{p}$ to a new 3D coordinate, while $\mathcal{T} : \mathbb{R}^3 \mapsto \mathbb{R}$ outputs the signed distance value at this new 3D coordinate.

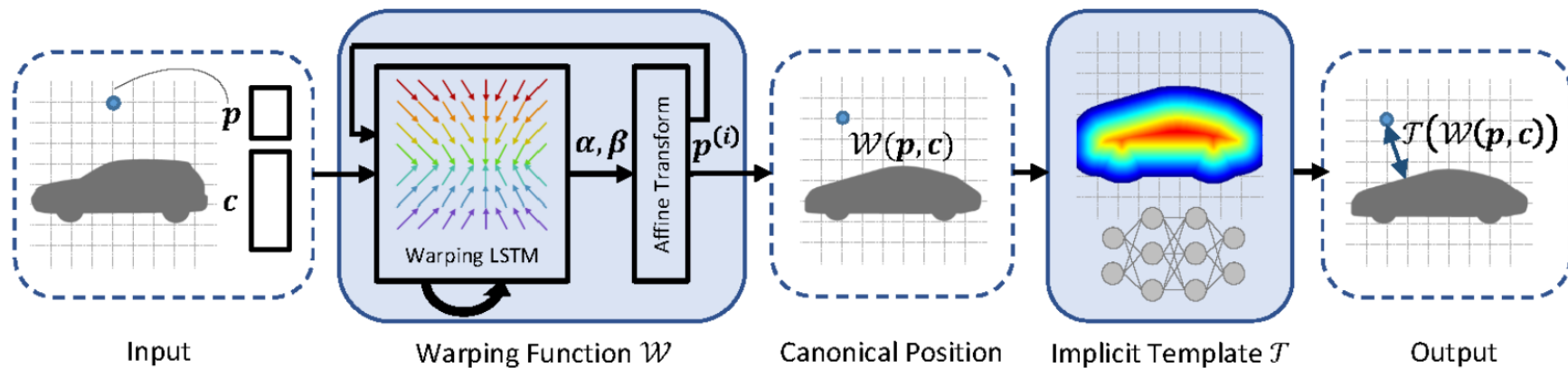


Figure 2: Method overview. Our method decomposes the DIF representation into a warping function and an implicit template. The warping function transforms point samples of shape c to their canonical positions, which are then mapped to SDF values by the implicit template.

- ◆ **Multiple steps of point-wise transformation using Warping LSTM**
- ◆ **different numbers of warping steps, outputs with s warping steps is**

$$\mathcal{L}_{rec}^{(s)} = \sum_{k=1}^K \sum_{i=1}^N L_{\epsilon_s, \lambda_s} \left(\mathcal{T} \left(\mathbf{p}^{(s)} \right), v_{k,i} \right) \quad (7)$$

- ◆ **Progressive reconstruction loss is the sum of all levels**

$$\mathcal{L}_{rec} = \sum_{s \in \{2,4,6,8\}} \mathcal{L}_{rec}^{(s)}$$

- ◆ 从一组形状中找出一个可信的模板
- ◆ 无监督实现密集的对对应关系

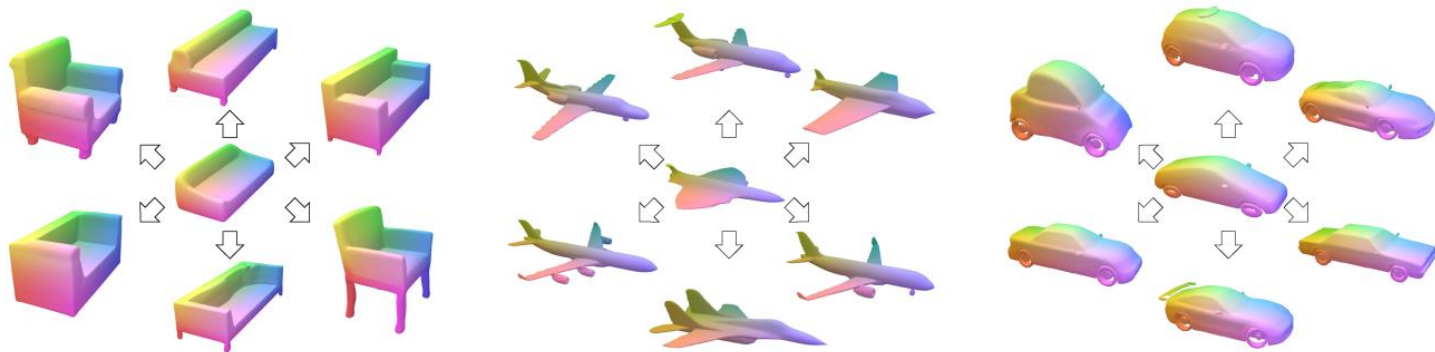
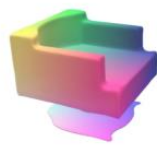


Figure 1: Example results of our representation. Our approach is able to factor out a plausible template (middle) from a set of shapes (surroundings), and builds up dense correspondences (color-coded) across all the shapes automatically without any supervision.



Chair Subclass #1



Chair Subclass #2



Table Subclass #1

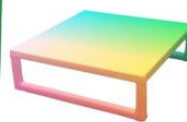
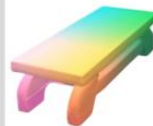


Table Subclass #2

Figure 4: More demonstration of our representation. For objects that show very high structure varieties, like chairs and tables, we can train individual deep implicit templates for different subclass. The learned templates are rendered in gray background and correspondences are shown with the same color.

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